

Business Objective

Problem Statements:

Strategically identify where to place 100 new DC fast chargers within a target state (Tennessee) to drive EV adoption and support long-term growth in the EV market.

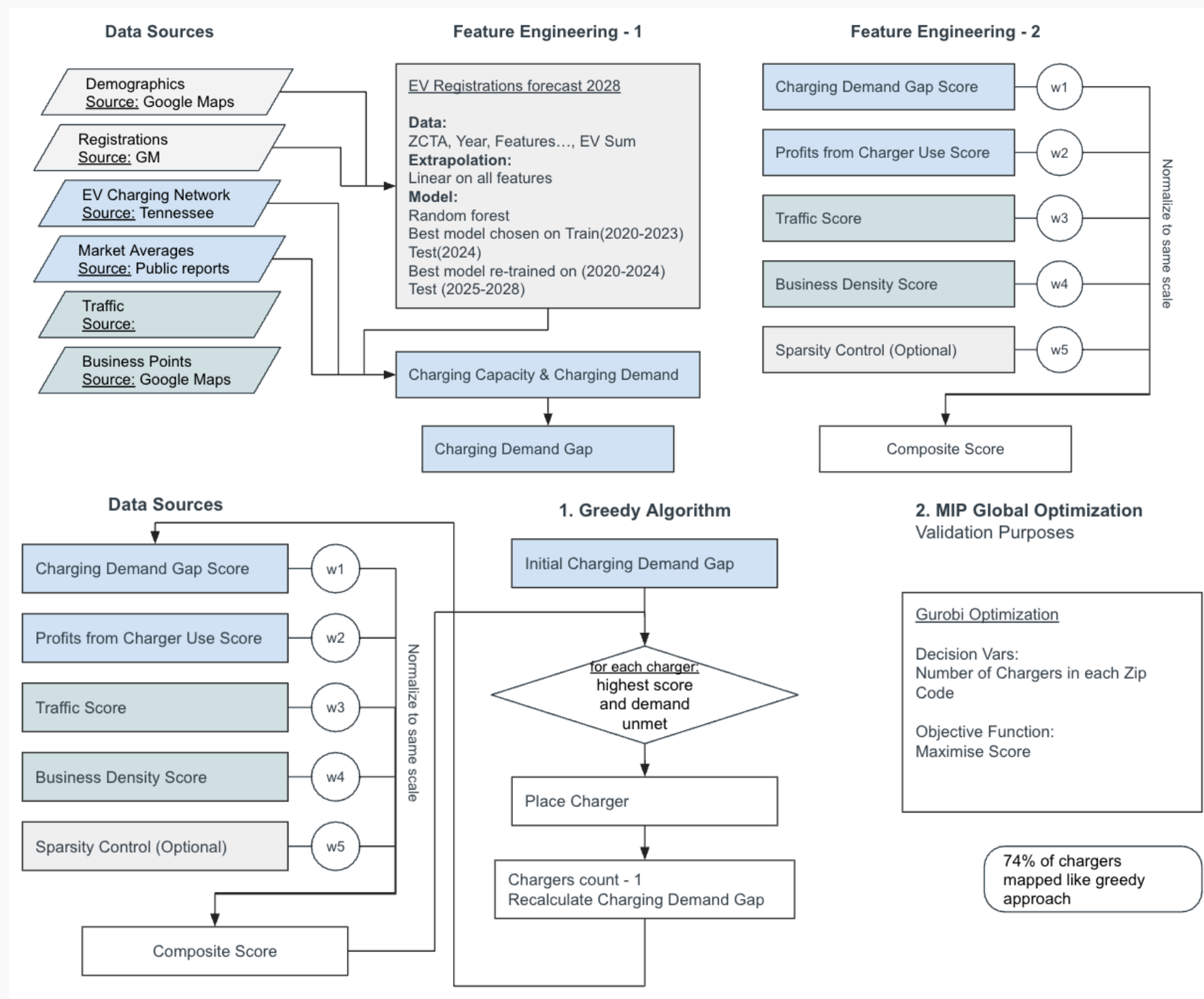
Expected Business Outcome:

1. Drive GM EV sales by addressing both current and forecasted gaps in public charging infrastructure
2. Support equitable EV adoption by expanding access across diverse regions

Data and Exploratory Insights

Data Descriptions: The dataset combines GM’s internal EV registration data with multiple open source datasets to provide a comprehensive view of EV readiness across Tennessee. It includes ZIP code level information on demographics, EV registration, EV charging network, business density, and traffic.

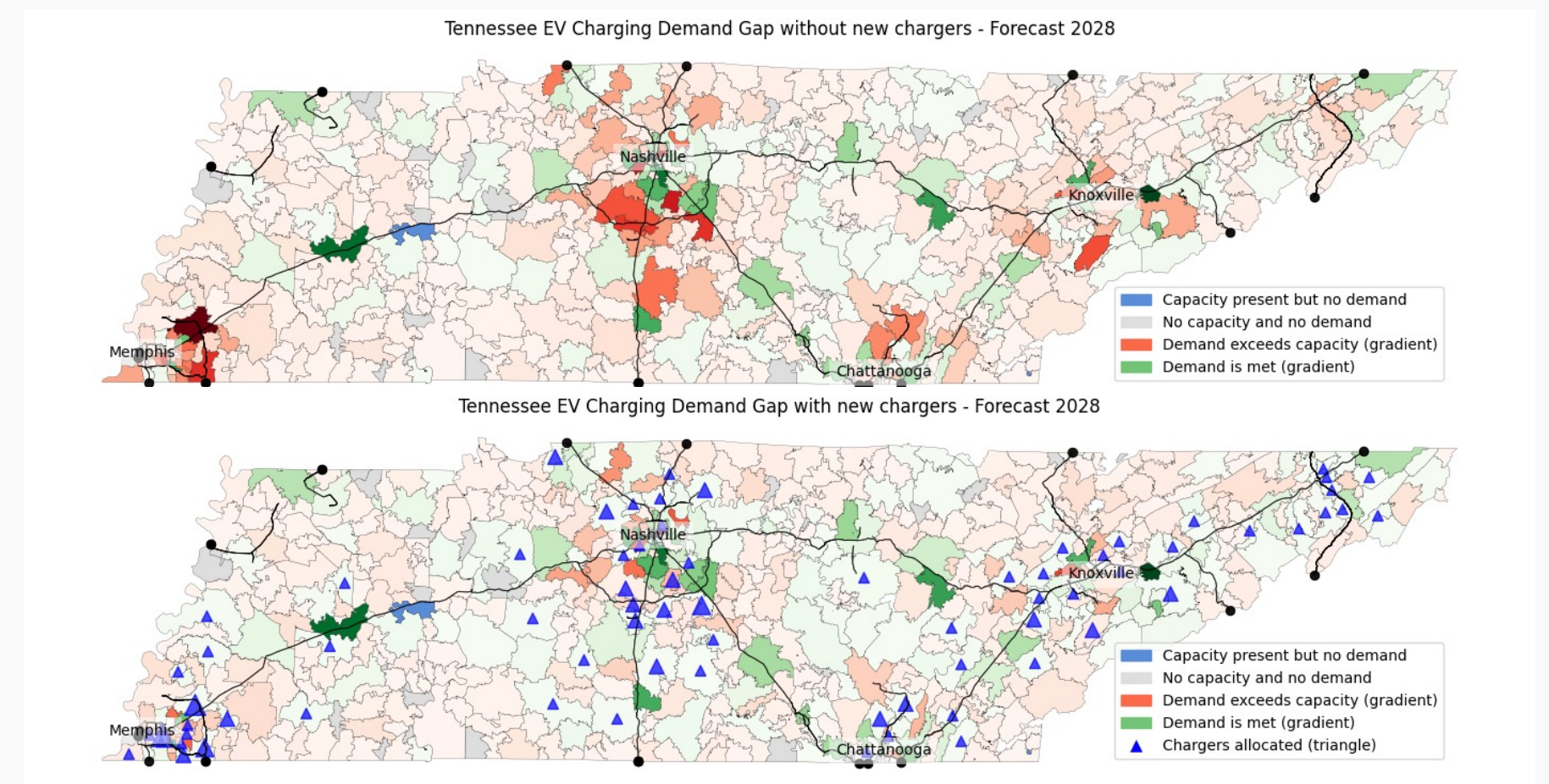
Data Workflow:



Methodology

Approaches	
1. Greedy Algorithm (Local Optimization)	Iteratively places chargers in the highest scoring ZIP codes where demand is still unmet. After each charger is placed, recalculate charging demand gap to adjust for shifting need.
2. Mixed-Integer Programming (Global Optimization)	Defines the number of chargers per ZIP as decision variables and seeks to maximize the total composite score , serving as a benchmark to validate our greedy approach

Model Result



Insights & Recommendations

- 1 **Greedy vs. Optimization:** Greedy algorithm enables fast, transparent, and scalable decision-making—ideal for real-world deployment
- 2 **Strategic placement reduces future gaps:** By allocating chargers based on our composite score, we significantly reduce projected demand gaps, especially in high-traffic corridors and population growth zones.
- 3 **EV Demand is growing faster than infrastructure:** Our 2028 demand forecasts revealed that several ZIP codes with rising EV adoption rates have no fast chargers today. This represent high-conversion opportunities where a single well-placed charger stimulate EV sales growth.